Abstract Harmonic Analysis

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The aim of this article is to review some basic concepts of *abstract harmonic analysis.*

1. Basic Theory of Banach Algebras

By a *normed algebra* (rsp. *Banach algebra*) is meant a normed linear space (rsp. Banach space) together with a multiplicative operation, which is compliant with the norm in some sense.

More specifically, let $(B, +, \circ)$ be a $\mathbb{K}$-algebra, that simultaneously constitutes a normed space $(B, \|\cdot\|)$. Then B is called a *normed algebra*, if the following conditions are fulfilled:

(A-1) $\|x \circ y\| \leq \|x\| \cdot \|y\|$ for each pair of elements $x, y \in B$.

(A-2) If B has an identity $\mathbf{1} \in B$, then $\|\mathbf{1}\| = 1$.

Condition (A-1) implies that the map $(x, y) \mapsto x \circ y$ is continuous. If (A-2) is valid, then B is called a *unital* algebra. Finally, B is said to be *commutative*, if

(A-3) $x \circ y = y \circ x$ for all $x, y \in B$.

As for normed linear spaces, completeness of normed algebras is of interest, thus a normed algebra is denoted a *Banach algebra*, if it is complete as a normed linear space.

The most prominent example of a non-commutative Banach algebra is given by the $\mathbb{C}$-algebra $\mathscr{L}(X)$ of bounded linear operators $A : X \to X$, where X is a real or complex Banach space. It forms a unital Banach algebra with respect to composition of mappings and the operator norm $\|A\|$ of $A \in \mathscr{L}(X)$.

One of the key results in the theory of complex Banach algebras is the *Gelfand-Mazur theorem* (see also [1], p. 14):

THEOREM .1. If each element $x \neq 0$ in a complex Banach algebra B is invertible, then $B \cong \mathbb{C}$.

In other words, theorem .1 quotes that the field of complex numbers is essentially the only complex Banach algebra, in which each element $x \neq 0$ is invertible.

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Let B be a complex and unital Banach algebra. By the *Neumann series* of $x \in B$ is meant the infinite sum $s_x := \sum_{n=0}^{\infty} x^n$. If s_x is convergent in B, then the element $\mathbf{1} - x$ is invertible, and a short computation proves $(\mathbf{1} - x)^{-1} = s_x$. Like in the case $B = \mathbb{C}$, the Neumann series is convergent for all $x \in B$ fulfilling $\|x\| < 1$.

If $x \in B$ is invertible, then any other element $x' \in B$ having the property

$$\|x - x'\| < \frac{1}{\|x^{-1}\|}$$

proves to be invertible, too. Thus all elements in a sufficiently small neighbourhood of an invertible element have convergent Neumann series and are invertible, which implies that the set $\mathbf{GL}(B)$ of invertible elements in B is open. It is called the *group of units* in B, since $x \circ y$ and x^{-1} are invertible for $x, y \in \mathbf{GL}(B)$. Both operations $(x, y) \mapsto x \circ y$ and $x \mapsto x^{-1}$ are continuous, so $\mathbf{GL}(B)$ forms a topological group, when considered in the subset topology of B.

Here are some important examples of commutative Banach algebras with respect to applications:

1. The field $\mathbb{C}$ of complex numbers may be regarded as a complex unital Banach algebra in its own right.
2. For any topological space X the function spaces $C_b(X)$ and $C_0(X)$, both endowed with the supremum norm $\|f\| := \sup_x |f(x)|$.
   If X is *compact*, then the function space $C(X)$ consisting of continuous functions $f : X \to \mathbb{C}$ forms a Banach algebra, which in addition is unital with identity $\mathbf{1}(x) := 1$ for $x \in X$.
3. The algebra $L^1(G, \mathcal{B}_G, \mu)$ of Lebesgue-integrable functions with respect to the Haar measure μ on a locally compact topological group G, where multiplication is defined by *convolution*

$$(f * g)(x) := \int_{\mathbb{R}} f(y)\, g(x - y)\, d\mu$$

   for $f, g \in L^1(G)$. Important examples for G are the classical groups $\mathbb{R}^n$, $\mathbb{Z}$ and $\mathbb{T} = \{z \in \mathbb{C} : \|z\| = 1\}$.
4. The Banach algebra $L^\infty(X)$ of essentially bounded functions $f : X \to \mathbb{R}$, where $(X, \mathcal{B}_X, \mu)$ may be any measure space. Addition and multiplication in this algebra are defined pointwise.
5. The algebra $\mathcal{M}^r(G)$ of regular complex Borel measures defined on a locally compact topological group G. The multiplication is

defined by convolution of measures:

$$(\mu * \nu)(B) \;:=\; (\mu \times \nu)(\{(x, y) \in G \times G : x + y \in B\})$$

for any Borel set $B \subset G$. This Banach algebra is even unital, since the so-called *Dirac measure* μ_0 owns the property $\mu_0 * \nu = \nu * \mu_0 = \nu$ for all $\nu \in \mathcal{M}^r(G)$.

2. Gelfand Representation

Given the Banach algebras B_1, B_2, a linear mapping $\Phi : B_1 \to B_2$ is denoted a *Banach algebra homomorphism*, if it preserves multiplication according to

$$\Phi(x \circ y) \;=\; \Phi(x) \circ \Phi(y) \text{ for all } x, y \in B_1.$$

Moreover, B_1 and B_2 are said to be *isomorphic* to each other, if there is a bijective homomorphism $\Phi : B_1 \to B_2$ fulfilling $\|\Phi(x)\|_{B_2} = \|x\|_{B_1}$ for all $x \in B_1$, which in this case is called a *Banach algebra isomorphism* (notation: $B_1 \cong B_2$).

The *Gelfand representation* is a powerful mapping in the setting of commutative Banach algebras and provides a Banach algebra homomorphism between an arbitrary commutative Banach algebra B and the algebra $C_0(X)$ of continuous functions vanishing at infinity, where X is assumed to be a locally compact Hausdorff space.

A *character* on a complex Banach algebra B is a *non-zero* Banach algebra homomorphism $\chi : B \to \mathbb{C}$. Characters (often also called *multiplicative functionals*) are of great importance in the theory of unital commutative Banach algebras, as we will see in the following.

By the multiplication property of a Banach algebra homomorphism, each character χ on B satisfyies the inequality $\|\chi\| \leq 1$, so χ is a continuous mapping and thus an element of the dual space B'. The set Γ of all characters is called the *structure space* or the *spectrum* of the Banach algebra B.

Due to the inequality $\|\chi\| \leq 1$ for all $\chi \in \Gamma$, the set Γ is a subset of the closed unit ball $B_1(0) \subset B'$. Thus we may endow Γ with the subset topology on $B_1(0)$ inherited from the weak-*-topology on B', which is called the *Gelfand topology* on the structure space. Now, by the *Alaoglu-Bourbaki theorem* the weak-*-topology on B' is weak*-compact, implying that the Gelfand topology must be locally compact and Hausdorff. Hence the structure space Γ becomes into a locally compact Hausdorff space. Moreover, if B is unital, it can be shown that the set Γ is even compact.

From now on let B be a *commutative* Banach algebra. We consider the natural mapping $\gamma : B \to C(\Gamma)$, where γ assigns to each $x \in B$ the function $\hat{x} \in C(\Gamma)$ by way of $\hat{x}(\chi) := \chi(x)$, that is, the function values of γ are given by evaluating the characters χ at the element x. The mapping γ is called the *Gelfand representation* of B.

We collect the main properties of the Gelfand representation for a commutative unital Banach algebra B to the following list:

1. γ is a Banach algebra homomorphism from B to $C(\Gamma)$ or $C_0(\Gamma)$, depending whether B is unital or not.
2. If B is unital, then $\hat{1}$ is the constant function 1 on Γ.
3. If B is unital, then $x \in B$ is invertible iff $\hat{x}$ nowhere vanishes.
4. $\|\hat{x}\|_\infty \leq \|x\|$ for each $x \in B$.

The Gelfand representation is injective iff B is a *semisimple* Banach algebra, that is, if the intersection of all maximal ideals in B, called the *Jacobsen radical* of B, is equal to the set $\{0\}$.

3. Generalities on Topological Groups

Before discussing abstract Fourier transformations, it is useful to review some concepts coming from general topology, especially that of a topological group, which plays a fundamental role in the theory of Fourier transforms.

There is a subclass within that of general topological spaces, which is of great importance in higher analysis and whose members are called *topological groups.* They are mathematical objects owning the algebraic structure of a group and a topological structure together. Both have to be compatible to each other in the following sense: let $(G, \circ)$ be an ordinary group and a topological space $(G, \mathcal{T})$ at once. Then G is called a *topological group*, if the group operation $(x, y) \mapsto x \circ y^{-1}$ is continuous.

Instead to require continuity of the mapping $(x, y) \mapsto x \circ y^{-1}$, one can equivalently require that the mappings $(x, y) \mapsto x \circ y$ and $x \mapsto x^{-1}$ have to be continuous.

Well-known examples of topological groups are:

1. each group G with its discrete topology 2^G,
2. the group $\mathbb{R}^\times := \mathbb{R} - \{0\}$,
3. the group $\mathbb{C}^\times := \mathbb{C} - \{(0, 0)\}$,

4. the group $\mathbb{R}^+ := \{x \in \mathbb{R} : x > 0\}$,
5. the additive groups $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$,
6. the *circle group* $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$,
7. each subgroup U of a topological group G endowed with the subspace topology and its closure $\overline{U}$ are topological groups.

Like for purely algebraic groups, a *homomorphism* between topological groups is defined to be a *continuous* group homomorphism, and such a homomorphism ι is called an *isomorphism*, if it is a group isomorphism and ι^{-1} is continuous, too (we use the notation $G \cong H$ in this case).

A well-known example of a group isomorphism is the *exponential function* $t \mapsto e^t$, which maps the topological group $(\mathbb{R}, +)$ to the group $(\mathbb{R}^+, \cdot)$. Rather similar, the function $t \in [0, 1) \mapsto e^{2\pi i t}$ forms a group isomorphism between the topological groups $\mathbb{R}/\mathbb{Z}$ and $\mathbb{T}$.

Because in a topological group G the left translation $x \mapsto g \circ x$ and the right translation $x \mapsto x \circ g$ are continuous mappings, they are both examples for *automorphisms*, that is, isomorphisms from the group G to itself. Hence the topological properties of a topological group are completely determined by the topology in a neighbourhood of the identity element $1 \in G$, which gives rise to call a topological group to be *homogeneous* in this case.

Moreover, the kind, how the identity 1 of a topological group G lies in a given subgroup U, determines the topology of that subgroup. In particular,

1. $U \subset G$ is open iff 1 is an inner point of U.
2. U is discrete iff 1 is an isolated point of U.

Remember that the circle group $\mathbb{T}$ is the group consisting of all complex numbers of modulus 1 with group operation

$$e^{i\alpha} \circ e^{i\beta} := e^{i\alpha} \cdot e^{i\beta} := e^{i(\alpha+\beta)}, \quad \alpha, \beta \in \mathbb{R}.$$

Endowed with the subset topology inherited from the standard topology on the field $\mathbb{C}$, the unit circle $\mathbb{T}$ forms a topological group. Consider the set $C(G, \mathbb{T})$ consisting of all continuous functions $f : G \to \mathbb{T}$. By furnishing this set with the compact-open topology, it becomes into a Hausdorff space.

A *character* χ is a (topological) group homomorphism from a topological group G to the circle group $\mathbb{T}$ and hence a continuous mapping. We denote G^* the set of characters $\chi : G \to \mathbb{T}$, which forms a subset of the Hausdorff space $C(G,\mathbb{T})$. We can make G^* into a topological space by taking the subset topology with respect to the compact-open topology on $C(G,\mathbb{T})$.

Performing multiplication of characters $\chi_1, \chi_2 \in G^*$ by

$$\chi_1 \circ \chi_2 : g \mapsto \chi_1(g) \cdot \chi_2(g),$$

we obtain a further character $\chi_1 \circ \chi_2 \in G^*$. It can be shown that this operation has all the properties of a group, and since it is commutative, G^* has the structure of an Abelian group, called the *character group* or the *dual group* of G. Moreover, the group operation in G^* is continuous, hence the dual group of any topological group forms a topological group as well.

In the following table we list three classical topological Abelian groups together with their dual groups and characters:

G	G^*	characters
$\mathbb{R}$	$\mathbb{R}^* \cong \mathbb{R}$	$\chi_x(t) := e^{itx}$, $x \in \mathbb{R}$
$\mathbb{Z}$	$\mathbb{Z}^* \cong \mathbb{T}$	$\chi_z(n) := e^{inz}$, $z \in \mathbb{T}$
$\mathbb{T}$	$\mathbb{T}^* \cong \mathbb{Z}$	$\chi_n(z) := z^n$, $n \in \mathbb{Z}$

Table 1: classical topological groups and dual groups.

Concerning cartesian products of topological groups, for each topological group G and H we have

$$(G \times H)^* \cong G^* \times H^*.$$

As a consequence, the group isomorphisms $\mathbb{R}^{n*} \cong \mathbb{R}^n$ hold for any integer n.

4. Locally Compact Abelian Groups

Recall that a topological space is called *locally compact*, if every subset owns a compact neighbourhood. The abbreviation *LCA group* usually stands for a *locally compact Abelian group*. The meaning of locally compact groups is based on the fact that these groups have non-trivial characters, that is, we have $G^* \neq \{\chi_1\}$ for such a group G, where χ_1 is the character with $\chi_1(x) = 1$ for all $x \in G$.

Suppose that the Abelian group G is *locally compact*. Then it can be shown (for example by the Arzelá-Ascoli theorem) that the dual group G^* is also locally compact with respect to its compact-open topology. Let G^{**} denote the *bidual group* of a topological group G, which is defined to be the dual group of its character group.

One of the main results in the theory of LCA groups is the famous *Pontryagin duality theorem*:

THEOREM .2. If G is LCA, then the groups G and G^{**} are isomorphic to each other.

Notice that each group element $g \in G$ gives rise to a character χ_x on the dual group G^* by $\chi_x(\xi) := \xi(x)$ for all $\xi \in G^*$. Thus the mapping $x \mapsto \chi_x \in G^{**}$ is an injective group homomorphism, and Pontryagin duality means that the canonical embedding $\iota : G \to G^{**}$, $x \mapsto \chi_x$ is also surjective.

It is easy to see that the groups listed in table 1 are examples of Pontryagin duality. This kind of duality also implies remarkable relationships between the topological properties of a LCA group G and its dual group G^*:

1. G is compact if and only if G^* is discrete.
2. G is discrete if and only if G^* is compact.

Finally, let us mention that it is possible to endow the convolution algebra $L^1(G, +, *)$ of any LCA group G with an involutional mapping $* : L^1(G) \to L^1(G)$, defined for $f \in L^1(G)$ by

$$f^*(x) := \overline{f(-x)}, \qquad (x \in G).$$

5. Abstract Fourier Transformation

Since $L^1(G, +, *)$ forms a commutative Banach algebra for every LCA group G, we may study its Gelfand transformation in detail, which in this case is denoted *Fourier transformation.* If Γ_G is the structure space of the algebra $L^1(G, +, *)$, then Fourier transformation maps each function $f \in L^1(G)$ to a function $\hat{f} \in C_0(\Gamma_G)$. It constitutes a Banach algebra homomorphism between $L^1(G)$ and $C_0(\Gamma_G)$, where convolution in $L^1(G)$ corresponds to multiplication in $C_0(\Gamma_G)$.

Moreover, each character $\chi \in G^*$ gives rise to a continuous multiplicative functional $\gamma \in \Gamma_G$, that is defined according to

$$\gamma(\chi) := \int_G f\,\overline{\chi}\,d\mu,$$

and reversely each continuous multiplicative functional $\gamma \in \Gamma_G$ can be represented in this way, Thus, we identify the structure space of the convolution algebra $L^1(G,+,*)$ with the dual group G^*, and Fourier transformation appears as the Gelfand transformation up to this identification:

1. For $f \in L^1(G)$ the mapping $\hat{f} : \widehat{G} \mapsto \mathbb{C}$ according to

$$\hat{f}(\chi) := \int_G f\,\overline{\chi}\,d\mu, \quad \chi \in \widehat{G}$$

   is called the *Fourier transform* of f.
2. The mapping $\mathscr{F} : f \in L^1(G) \to \hat{f} \in C_0(G^*)$ is called *Fourier transformation.*

Let us repeat some properties of Fourier transformation that are already known from the general Gelfand transformation:

1. For every $f \in L^1(G)$ the mapping $\hat{f}$ is continuous on G^*.
2. the function $\hat{f}$ is vanishing at infinity (*Riemann-Lebesgue theorem*), that is, it maps absolutely integrable functions to elements of the function space $C_0(G^*)$.
3. $\mathscr{F}$ is injective, that is, $\mathscr{F}(f) \neq \mathscr{F}(g)$ for $f \neq g$.
4. $\mathscr{F}$ is norm-preserving, that is, $\|\mathscr{F}(f)\|_\infty \leq \|f\|_1$.
5. $\mathscr{F}(\alpha f + \beta g) = \alpha \mathscr{F}(f) + \beta \mathscr{F}(g)$, $\alpha, \beta \in \mathbb{R}$.
6. $\mathscr{F}(f * g) = \sqrt{2\pi}\,\mathscr{F}(f) \cdot \mathscr{F}(g)$.

In addition, the image of $L^1(G)$ under $\mathscr{F}$ forms a separating subalgebra of $C_(\Gamma_G)$, which lies even dense in $C_0(\Gamma_G)$.

It is nearby to consider Fourier transformation on $L^1(G,+,*)$ for the classical groups $\mathbb{R}$, $\mathbb{R}^n$, $\mathbb{Z}$ and $\mathbb{T}$:

1. If $G = \mathbb{R}$, then the character group of $(\mathbb{R},+)$ is the group $(\mathbb{R},+)$ itself. Hence a function $f \in L^1(\mathbb{R})$ owns the Fourier transform $\hat{f} : \mathbb{R} \to \mathbb{C}$, whose function values are defined as follows:

$$\hat{f}(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)\,e^{-itx}\,dt, \quad (x \in \mathbb{R}).$$

2. In case of $G = \mathbb{R}^n$ the character group of $(\mathbb{R}^n,+)$ coincides with the group $(\mathbb{R}^n,+)$ as well. The Fourier transform of $f \in L^1(\mathbb{R}^n)$

is the function defined by

$$\hat{f}(x) := \frac{1}{(2\pi)^{n/2}} \int_{-\infty}^{\infty} f(t)\, e^{-i\langle t,x\rangle}\, dt, \quad (x \in \mathbb{R}^n).$$

3. Let $G = \mathbb{Z}$. The normed Haar measure on this group is nothing but the counting measure, and the character group $(\mathbb{Z}, +)$ is isomorphic to the multiplicative group $(\mathbb{T}, \cdot)$. Hence the Fourier transform of a sequence $\{f_n\}_{n\in\mathbb{N}}$ is the mapping $\hat{f} : \mathbb{T} \to \mathbb{C}$, which is defined pointwise by the *Fourier series*

$$\hat{f}(\omega) := \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} f_n\, e^{-in\omega}, \quad (\omega \in \mathbb{T}).$$

4. The character group of the group $(\mathbb{T}, \cdot)$ is isomorphic to the group $(\mathbb{Z}, +)$, hence the Fourier transform of a function $f : \mathbb{T} \to \mathbb{R}$ is the sequence $\hat{f} : \mathbb{Z} \to \mathbb{C}$ with members

$$\hat{f}_n := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{T}} f(z)\, e^{-inz}\, dz, \quad (n \in \mathbb{Z}).$$

By *inverse Fourier transformation* is meant the procedure, how certain functions in the algebra $L^1(G)$ can be retrieved from the function values of the corresponding Fourier transforms.

Since the adjoint group G^* of a LCA group G with Haar measure μ is LCA again, it owns a Haar measure as well, hence the convolution algebra $L^1(G^*)$ is well-defined. Let f be an element of $L^1(G)$ fulfilling the additional condition $\hat{f} \in L^1(G^*)$. Then there exists the Fourier transform of $\hat{f}$,

$$\hat{\hat{f}}(x) = \int_{G^*} \hat{f}(\gamma)\overline{x(\gamma)}\, d\nu,$$

which forms a complex-valued mapping defined on the biadjoint group G^{**}. Due to Pontryagin duality, G^{**} is isomorphic to G, and we may define the *inverse Fourier transform* F according to

$$F(x) = \int_{G^*} \hat{f}(\gamma)\, \overline{\gamma(x)}\, d\mu, \quad (x \in G).$$

It can be shown that the functions f and F are μ-nearly everywhere identical on G.

Especially for the group $(\mathbb{R},+)$, where even $\mathbb{R}^*$ may be identified with $\mathbb{R}$, we obtain the inverse Fourier transform of $\hat{f} \in L^1(\mathbb{R})$ according to

$$F(x) \;=\; \int_R \hat{f}(t)\, e^{itx}\, dt, \quad (x \in \mathbb{R}).$$

Let X be any topological space and μ be a measure define on the Borel σ-algebra $\mathcal{B}_X$. A Borel set $B \subseteq X$ is called *inner regular* if

$$\mu(B) \;=\; \sup\{\mu(K) : K \subset B,\, K \text{ kompact}\}.$$

Then a measure μ is denoted as *inner regular*, if all Borel sets are so. Hence inner regular measures defined on X are already uniquely defined by their values for compact subsets of X. Each inner regular measure on a topological space is said to be *Radon* if it adopts finite values for compact sets.

Rather similar, a measure μ is called *outer regular*, if for each Borel set $B \subseteq X$ the equality

$$\mu(B) \;=\; \inf\{\mu(\mathcal{O}) : B \subset \mathcal{O},\, \mathcal{O} \text{ open}\}.$$

holds. Finally, a measure is called *regular*, if it is inner and outer regular at the same time.

Let $\mathscr{M}_r(G)$ be the Banach space of finite regular Borel measures, where G is a LCA group. Let μ_1, μ_2 be measures in $\mathscr{M}_r(G)$ and let $\mu_1 \times \mu_2$ be the product measure defined on the group $G \times G$. If $B \in \mathcal{B}_G$ is measurable, then the set B' is defined by

$$B' \;:=\; \{(x,y) \in G \times G : x + y \in B\}.$$

This set is measurable with respect to the σ-Algebra in $G \times G$. Moreover, the *convolution* operation $\mu_1 * \mu_2$ of two measures μ_1 und μ_2 is defined to be the measure

$$(\mu_1 * \mu_2)(B) \;:=\; (\mu_1 \times \mu_2)(B') \quad (B \in \mathcal{B}_G),$$

which owns the following properties:

1. $\mu_1 * \mu_2 \in \mathscr{M}_r(G)$,

2. $(\mu_1, \mu_2) \mapsto \mu_1 * \mu_2$ is commutative and associative,
3. the inequality $\|\mu_1 * \mu_2\| \leq \|\mu_1\| \cdot \|\mu_2\|$ holds.

Hence the Banach algebra $\mathscr{M}_r(G, +, *)$ is unital and commutative. Its unit element is the *Dirac measure* δ_0, which distinguishes from other measures in this algebra by the property that it is concentrated to the element $0 \in G$.

In case of $G = (\mathbb{R}^n, +)$, we may regard the algebra $\mathscr{M}_r(\mathbb{R}^n, +, *)$ as the Banach algebra, that is norm-isomorphic to the completion of the non-unital convolution algebra $L^1(\mathbb{R}^n, +, *)$.

Fourier-Stieltjes transformation is another phrase for Gelfand transformation on the unital and commutative Banach algebra $\mathscr{M}_r(G, +, *)$ of finite and regular complex Borel measures on a LCA group G. In fact, if $\mu \in \mathscr{M}_r(G, +, *)$, then the mapping $\hat{\mu}$ defined on the dual group G^* by

$$\hat{\mu}(\gamma) := \int_G -\gamma(x)\, d\,\mu(x), \qquad \gamma \in G^*$$

is called *Fourier-Stieltjes transform* of μ.

Let $B(G^*)$ be the set of functions $\hat{\mu}$ with $\mu \in \mathscr{M}_r(G, +, *)$. Then the following properties are valid:

1. Each function $\hat{\mu} \in B(G^*)$ is bounded and uniformly continuous.
2. If $\sigma = \mu * \nu$, then $\hat{\sigma} = \hat{\mu} \cdot \hat{\nu}$ holds, that is, the mapping $\mu \mapsto \hat{\mu}(\gamma)$ is a Banach algebra endomorphism on $\mathscr{M}_r(G, +, *)$ for every $\gamma \in X^*$.
3. The set $B(G^*)$ is invariant under translation, multiplication and complex conjugation.

6. Fourier Transformation in Euclidean Spaces

To the end we elaborate *Fourier transformation* in the spaces $\mathbb{R}^d$, which is an important tool in advanced analysis with many useful applications, especially to the study of linear differential operators with constant coefficients.

Rather Fourier transformation can be defined in various linear spaces, it is customary to introduce this mapping first in Lebesgue space $L^1(\mathbb{R}^d)$. Recall that this space contains all measurable functions $f : \mathbb{R} \to \mathbb{C}$ having the property

$$\int_{\mathbb{R}^d} |f|\,dx \;<\; \infty.$$

Preliminary, for elements $f, g \in L^1(\mathbb{R}^d)$ a multiplication $f * g$ is defined by

$$(f * g)\,(x) \;:=\; \int_{\mathbb{R}^d} f(t) \cdot g(x-y)\,dy$$

and denoted the *convolution* in $L^1(\mathbb{R}^d)$. This mapping is commutative, associative and fulfills for all $f, g \in L^1(\mathbb{R}^d)$ the estimate

$$\|f * g\|_1 \;\leq\; \|f\| \cdot \|g\|\,.$$

Since $L^1(\mathbb{R}^d)$ is complete, the algebra $(L^1(\mathbb{R}^d), +, *)$ constitutes a commutative but non-unital Banach algebra.

L1-Fourier transformation in Euclidean space $\mathbb{R}^d$ is nothing but the Gelfand mapping $\mathscr{F} : f \to \hat{f} \in C_0(\mathbb{R}^d)$, which sends each function $f \in L^1(\mathbb{R}^d)$ to the function $\hat{f} \in C_0(\mathbb{R}^d)$, called the *Fourier transform* of f and defined by

$$\hat{f}(\xi) \;:=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-2\pi i \langle \xi, x\rangle}\, f(x)\,dx \qquad (\xi \in \mathbb{R}^d), \tag{6.1}$$

where $\langle \xi, x\rangle$ is the usual inner product of $\xi, x \in \mathbb{R}^d$.

Let us again collect the remarkable properties of the mapping $\mathscr{F}$ in the special case $G = G^* = \mathbb{R}^d$:

1. For each $f \in L^1(\mathbb{R}^d)$ the function $\hat{f}$ is continuous on $\mathbb{R}^d$, so $\hat{f} \in C(\mathbb{R}^d)$.
2. $\hat{f}$ is vanishing at infinity, so $\hat{f} \in C_0(\mathbb{R}^d)$.
3. $\mathscr{F}$ is injective, that is, $\mathscr{F}(f) \neq \mathscr{F}(g)$ for $f \neq g$.
4. $\|\mathscr{F}(f)\|_\infty \leq \|f\|_1$ for $f \in L^1(\mathbb{R}^d)$.
5. $\mathscr{F}(\alpha f + \beta g) = \alpha\, \mathscr{F}(f) + \beta\, \mathscr{F}(g)$ for $\alpha, \beta \in \mathbb{C}$, thus $\mathscr{F}$ is linear.
6. $\mathscr{F}(f * g) \;=\; \mathscr{F}(f) \cdot \mathscr{F}(g)$, so $\mathscr{F}$ is an algebra homomorphism between the Banach algebras $L^1(\mathbb{R}^d, +, *)$ and $C_0(\mathbb{R}^d, +, \cdot)$.
7. If $f \in L^1(\mathbb{R}^d)$ and $g : x \in \mathbb{R}^d \mapsto |x| f(x) \in L^1(\mathbb{R}^d)$, then $\hat{f} \in C^1(\mathbb{R}^d)$ and
$$\partial_j f(x) \;=\; -\widehat{i\,\xi_j f(\xi)} \quad (j = 1, \ldots, d).$$

We now introduce Fourier transformation in the space $L^2(\mathbb{R}^d)$. Indeed, Recall that this space consists of the measurable functions $u:\mathbb{R}^d \to \mathbb{C}$ fulfilling

$$\int_{\mathbb{R}^d} |u|^2\, dx \;<\; \infty,$$

and the inner product in $L^2(\mathbb{R}^d)$ is given by

$$\langle u, v\rangle \;=\; \int_{\mathbb{R}^d} u\,\overline{v}\, dx.$$

If the functions $u, v:\mathbb{R}^d \to \mathbb{C}$ are Lebesgue-integrable then by definition they belong to the space $L^1(\mathbb{R}^d)$, and their *Fourier transforms* are well-defined by formula (6.1).

However, if is not only a member of $L^1(\mathbb{R}^d)$, but also $u \in L^2(\mathbb{R}^d)$, we define Fourier transformation a little bit different:

$$\hat{u}(\xi) \;:=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-2\pi i\langle \xi, x\rangle}\, u(x)\, dx. \tag{6.2}$$

Then for functions $u, v \in L^2(\mathbb{R}^d)$ the inner product between u and v and the inner product of their Fourier transforms $\hat{u}$ and $\hat{v}$ are related by PLANCHEREL*'s formula*

$$\langle u, v\rangle \;=\; \langle \hat{u}, \hat{v}\rangle.$$

Hence *Fourier transformation* $\mathscr{F}: u \mapsto \hat{u}$ constitutes a partial isometry from the space $L^{1,2}(\mathbb{R}^d) \;:=\; L^1(\mathbb{R}^d) \,\cap\, L^2(\mathbb{R}^d)$ to the space $L^2(\mathbb{R}^d)$.

It is a crucial fact now that $L^{1,2}(\mathbb{R}^d)$ proves to be dense in $L^2(\mathbb{R}^d)$. Since the mapping $\mathscr{F}$ is linear and continuous on $L^{1,2}(\mathbb{R}^d)$, it is even *uniformly continuous* on that space and can be extended to an *unitary*, that is, a bijective and isometric linear operator $\mathscr{F}_P: L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$, called the FOURIER-PLANCHEREL *operator* or the L^2*-Fourier transformation* in $L^2(\mathbb{R}^d)$.

But in general a function $u \in L^2(\mathbb{R}^d)$ is not an element of $L^1(\mathbb{R}^d)$. Thus it is not possible to define the L^2-Fourier transform $\hat{u}$ of every $u \in L^2(\mathbb{R}^d)$ by means of formula (6.2). However, for $u \in L^2(\mathbb{R}^d)$ and $R > 0$ the function

$$\hat{u}_R(\xi) \;:=\; \frac{1}{(2\pi)^{d/2}} \int_{|x|\le R} e^{-2\pi i\langle x, \xi\rangle}\, u(x)\, dx$$

is well-defined, and one may obtain the L^2-Fourier transform $\hat{u}$ as the limit of the functions $\hat{u}_R$ for $R \to \infty$ with respect to the 2-norm, that is, $\lim_{R\to\infty} \|\hat{u} - \hat{u}_R\|_2 = 0$.

The Fourier-Plancherel operator can be used to introduce the class of *Bessel potential spaces.* First recall that the mapping $\mathscr{F}_P$ is well-defined in Hilbert space $L^2(\mathbb{R}^d)$, therefore it has also a meaning in the linear subspaces $H^k(\mathbb{R}^d)$ for $k \in \mathbb{N}$. We define for each $k \in \mathbb{N}$ the real-valued mapping

$$p_k : x \in \mathbb{R}^d \mapsto (1 + |x|^{2k})^{1/2} \tag{6.3}$$

and the space

$$\mathcal{H}^k(\mathbb{R}^d) := \{f \in L^2(\mathbb{R}^d) : p_k \hat{f} \in L^2(\mathbb{R}^d)\},$$

which is Hilbertian with inner product $\langle f, g\rangle_{p_k} := \langle p_k \hat{f}, p_k \hat{g}\rangle$ for $f, g \in \mathcal{H}^k(\mathbb{R}^d)$ and derived norm

$$\|f\|_{p_k}^2 := \langle f, f\rangle_{p_k} = \int_{\mathbb{R}^d} (1 + |x|^{2k})\, |\hat{f}|^2 \, dx.$$

It turns out that the so-called *Bessel potential space* $\mathcal{H}^k(\mathbb{R}^d)$ is the range of the FOURIER-PLANCHEREL operator $\mathscr{F}_P$ when restricted to $H^k(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$. Hence $\mathscr{F}_P$ is a bijective and norm-preserving mapping from $H^k(\mathbb{R}^d)$ to $\mathcal{H}^k(\mathbb{R}^d)$, i. e.

$$|f|_k = \|\hat{f}\|_{p_k} \qquad \text{for all } f \in H^k(\mathbb{R}^d).$$

In order to extend the definition of the spaces $H^k(\mathbb{R}^d)$ to indices $s \in \mathbb{R}^+$, let p_s be defined as in (6.3) but with $s > 0$ instead of $k \in \mathbb{N}$. We define

$$\mathcal{H}^s(\mathbb{R}^d) := \{\, f \in L^2(\mathbb{R}^d) : p_s \hat{f} \in L^2(\mathbb{R}^d)\,\},$$

then $\mathcal{H}^s(\mathbb{R}^d)$ given this way coincides for $s = k \in \mathbb{N}$ with the classical Sobolev space $H^k(\mathbb{R}^d)$, which motivates to define $H^s(\mathbb{R}^d) := \mathcal{H}^s(\mathbb{R}^d)$ for $s \notin \mathbb{N}$. For $s < 0$, however, we introduce $H^s(\mathbb{R}^d) := H^{-s}(\mathbb{R}^d)^*$.

We finally elaborate that Fourier transformation can be extended to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions.

First notice that the Fourier transform $\hat{\phi}$ of a Schwartz function $\phi \in \mathscr{S}(\mathbb{R}^d)$ is well defined, since $\mathscr{S}(\mathbb{R}^d)$ is a linear subspace of the convolution algebra $L^1(\mathbb{R}^d, +, *)$. Moreover, Fourier transformation restricted to the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ forms a linear mapping $\mathscr{F}: \mathscr{S}(\mathbb{R}^d) \to \mathscr{S}(\mathbb{R}^d)$, which sends each function $\phi \in \mathscr{S}(\mathbb{R}^d)$ to a further function $\hat{\phi} \in \mathscr{S}(\mathbb{R}^d)$ according to the well-known formula

$$\hat{\phi}(x) = \mathscr{F}(\phi)(x) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i<\xi,x>} \phi(\xi)\, d\xi \quad (x \in \mathbb{R}^d). \quad (6.4)$$

The extraordinary meaning of Schwartz functions is founded by the fact that the Fourier transformation has excellent properties on the space $\mathscr{S}(\mathbb{R}^d)$. More precisely, besides the well-known features of Fourier transformation in $L^1(\mathbb{R}^d, +, *)$, the mapping $\mathscr{F}$ is *bijective* on $\mathscr{S}(\mathbb{R}^d)$. Hence the *inverse Fourier transform* exists for all $\phi \in \mathscr{S}(\mathbb{R}^d)$ and is described by the *Fourier inversion formula*

$$\check{\phi}(\xi) = \mathscr{F}^{-1}(\phi)(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle} \phi(x)\, dx \qquad (\xi \in \mathbb{R}^d).$$

We next extend Fourier transformation in $\mathscr{S}(\mathbb{R}^d)$ to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions by transposition, that is

$$(\mathscr{F}T)\,\phi := T(\mathscr{F}\phi), \qquad \phi \in \mathscr{S}(\mathbb{R}^d),\ T \in \mathscr{S}'(\mathbb{R}^d).$$

This mapping is bijective in the space $\mathscr{S}'(\mathbb{R}^d)$ and even continuous with respect to the weak*-topology on $\mathscr{S}'(\mathbb{R}^d)$. The inverse Fourier transform of a tempered distribution T is represented by the formula

$$\mathscr{F}^{-1}(T)(\phi) := T(\mathscr{F}^{-1}\phi) \qquad (\phi \in \mathscr{S}(\mathbb{R}^d)).$$

For example, the Fourier transform $\mathscr{F}(\delta_0)$ of the Dirac distribution $\delta_0 \in \mathcal{S}'(\mathbb{R})$ is the continuous linear functional

$$\mathscr{F}(\delta_0)(\phi) := \delta_0(\mathscr{F}(\phi)) = \hat{\phi}(0) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \phi(x)\, dx, \quad \phi \in \mathscr{S}(\mathbb{R}),$$

so $\mathscr{F}(\delta_0)$ is nothing but the distribution induced by the constant function $x \mapsto \frac{1}{\sqrt{2\pi}}$.

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